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Cohesiveness reduces foraging efficiency in a social herbivore

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For social foragers, movement as a group could increase foraging efficiency through collective discovery of high-quality food sources. This would require an efficient mechanism for transferring information about food quality between individuals. Conversely, the constraints of foraging as a cohesive group could decrease efficiency; grouping may persist to serve other functions such as protection from predators. To test what drives cohesion in herbivores, we manipulated patch shape and within-patch pattern of food quality and quantified the effects on group level diet selection by a social herbivore, the fallow deer, *Dama dama*. We arranged feeders containing fodder in lines or blocks, and manipulated the pattern of food quality within patches by adding tannin, a plant secondary compound that decreases palatability. We quantified the relative consumption of low- and high-tannin food to compare diet selectivity at the group level between patch treatments. If group foraging evolved to increase foraging efficiency, altering the spatial arrangement of food should not affect diet selectivity because information about food location and quality is shared. We found, however, that the herd expressed different levels of selectivity between both patch shapes and food quality patterns. Deer selected better diets in blocks than lines. In lines, the herd selected better diets when quality varied between alternate feeders rather than between the two halves of the patch, suggesting a reliance on personal rather than group information. Deer consumed the most at patch centres in all treatments except in blocks with high-tannin centres, but diet selection was poorer in the latter compared to blocks with low-tannin centres. Aggregation at the centre of patches appears to have restricted exploitation of the best food. Predation pressure and/or resource variability may have favoured the evolution of a foraging strategy that prioritizes social cohesion over effective diet selection.
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Foraging is a critical aspect of an animal's life that ultimately affects its fitness. For most large generalist herbivores, the optimal foraging strategy involves selecting relatively high-quality food from among abundant low-quality food (Westoby, 1974). Quality, and thus preference, is defined by the nutrient content, chemical and physical defences and morphology of plants, as well as by the nutritional needs and detoxification abilities of herbivores at a particular point in time (Freeland & Janzen, 1974; Kimball & Provenza, 2003; McArthur, Hagerman, & Robbins, 1991). It is further modified by the availability of foods, which may change seasonally and with environmental disturbances (Di Stefano & Newell, 2008; Shipley, Blomquist, & Danell, 1998). Herbivores must therefore negotiate a mosaic of food quality that is both temporally and spatially variable. The expression of these preferences often results in the consumption of plants in proportions differing from their availability, and thus diet selection is important not only from the herbivore perspective but also because it can alter the composition of plant communities (Augustine & McNaughton, 1998).

Understanding diet selection by large herbivores is, however, complicated by a plethora of factors external to plant quality. Herbivores must make trade-offs between the benefits of selecting the highest quality plants and the costs incurred to do so. These costs include the time and energy invested in searching for and assessing the quality of plants relative to the available vegetation, involving the use of visual and olfactory cues, or direct sampling of plants (Fortin, 2003; Freidin & Kacelnik, 2011; Stutz, Banks, Dexter, & McArthur, 2015; Stutz, Croak, Proschogo, Banks, & McArthur, 2017). How efficiently herbivores detect and choose between available plants depends not only on their sensory abilities but also on the spatial distribution of plants (Bell, 1990; Bergvall, Rautio, Sirén, Tuomi, & Leimar, 2008; Etzenhouser, Owens, Spalinger, & Murden, 1998; Kotliar & Wiens, 1990). It follows that investment

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in search and assessment behaviours would require either reduced quantitative intake or more time spent foraging at the cost of other activities. Importantly, there is mounting evidence across a variety of taxa for a trade-off between diet selectivity and behaviours that minimize predation risk (Lima & Dill, 1990). In some cases, foragers accept greater perceived risk to obtain higher quality food, while in other scenarios, predator-induced vigilance has resulted in reduced selectivity between food items, presumably depending on the costs and benefits of predator avoidance and selectivity (Lima, 1988; Lima & Valone, 1986; McArthur, Orlando, Banks, & Brown, 2012; Metcalfe, Huntingford, & Thorpe, 1987). Suboptimal diet selection can therefore result when the cost of maximizing plant quality outweighs the benefits.

Animals that forage in groups can reduce some of the costs of diet selection compared to their solitary counterparts. Group-living animals can enhance their foraging efficiency by using social information from conspecifics, such as the location and quality of a resource patch (Dall, Giraldeau, Olsson, McNamara, & Stephens, 2005; Rieucau & Giraldeau, 2011). Individuals may be attracted to a patch due to the presence of conspecifics (local enhancement; Pöysä, 1992), and the outcomes of foraging and behavioural decisions of other individuals can inform about patch quality once there (public information; Valone & Templeton, 2002). This information can be efficiently transferred without using specific signals or mutual recognition of informed and naïve individuals (Couzin, Krause, Franks, & Levin, 2005). In addition, shared vigilance can reduce individual investment in vigilance behaviour, freeing up more time to forage (Dias, 2006; Elgar, 1989). Foraging in groups can also provide protection from predation via earlier detection of potential predators (Powell, 1974; Pulliam, 1973), reduced individual predation risk via the dilution effect (Foster & Treherne, 1981), predator confusion (Milinski, 1984), and the use of other group members as buffers from predators (Hamilton, 1971). Any of these mechanisms could lead an individual to perceive a lower risk of predation when foraging among conspecifics.

Inevitably, group foraging is also associated with costs. Individuals in groups may experience reduced foraging efficiency in terms of both the amount and quality of food ingested via competition and interference with conspecifics (Giraldeau, Valone, & Templeton, 2002; Krause & Ruxton, 2002; Valone, 1993). As group size increases, there is less food available per individual foraging in a patch and it is likely that some will be forced to eat lower quality food. Under competitive conditions, individuals may sacrifice selectiveness to maximize intake of a depleting resource. Unless the resources in a patch are abundant, the costs of sharing them are rarely offset by a heightened ability to find resource patches because individual search areas will overlap; it is, however, likely to reduce individual variance in intake (Ruxton, 1995). Interference between individuals foraging in a group can lead to the consumption of poorer diets by restricting the ability to move about in a patch and increasing aggressive interactions and theft of discovered food (Free, Beddington, & Lawton, 1977; Molvar & Bowyer, 1994). The effects of competition and interference thus limit the freedom of an individual to choose food with high quality within a mosaic of abundant low-quality food.

Use of shared information can also be detrimental to diet selection. Poor decisions by a single individual can be perpetuated throughout the group; it may not be possible to use personal and social information simultaneously, so observers may only be informed by behavioural decisions (not the cue used by the observed) leading to informational cascades (Giraldeau et al., 2002). Such processes may decrease individual fitness, not only as a result of the poor diet selection but also as a result of increased direct mortality due to predation. For example, by following conspecifics experienced with agricultural patches, bison, *Bison bison*, increased their exposure to hunters, resulting in a dramatic population decline in Prince Albert National Park, Canada (Sigaud et al., 2017). In addition, herbivores may choose lower quality food patches or leave good patches early to stay together (Dumont & Boissy, 2000; Scott, Provenza, & Banner, 1995; Valone, 1993). To avoid this, individuals must effectively balance interdependence and independence (List, Elsholtz, & Seeley, 2009). If interdependence is too low, animals fail to reach consensus, and group cohesion and behavioural synchronization can be weak, resulting in the group splitting up (e.g. subgroups of sheep, *Ovis aries*; Howery, Provenza, Banner, & Scott, 1996; Roath & Krueger, 1982). On the other hand, if independence is too low, choices can be suboptimal (List et al., 2009). Relatively little is known about how the costs of competition when foraging in a group interact with group cohesiveness, even though these costs lie at the root of group splitting.

Few empirical studies have examined the influence of collective behaviour and spatial pattern of food on foraging decisions in a group-living mammalian herbivore. Here, we used the fallow deer, *Dama dama*, as a model organism to test how the shape of a food patch and the distribution of food quality within it affect group level diet selection by a herd of social foragers. Fallow deer are appropriate models for testing this because their foraging behaviour is known to be influenced by both social factors and the spatial arrangement of food. Fallow deer herds are nonpermanent units that split up and fuse, with larger groups showing reduced foraging success (Focardi & Pecchioli, 2005; Gerard, Bideau, Maublanc, Loisel, & Marchal, 2002). Documented patterns of feeding behaviour by fallow deer herds suggest that they prioritize cohesion over food quality. Consumption of nonpreferred Norway spruce, *Picea abies*, probably by lower-ranked peripheral individuals, was greater in close proximity to supplemental feeding sites with high-quality food (Garrido, Lindqvist, & Kjellander, 2014). Also, in an experiment using feeders of high- and low-tannin pellets, fallow deer foraging in groups consumed more tannins than fallow deer foraging alone; that is, they were less selective in groups (Bergvall, Rautio, Kesti, Tuomi, & Leimar, 2006). Thus, group cohesion seems to be a priority for fallow deer, probably because nonforaging functions such as reduced predation risk outweigh the costs of foraging together (Beecham & Farnsworth, 1999). How the prioritization of cohesion interacts with the spatial distribution of food resources is not well understood.

To test the group level feeding response of fallow deer to patch shape, we used bowls of pelleted food (feeders) arranged in lines and blocks to create elongated and compact patches. In a scenario where information is transmitted efficiently between individuals and there are no competing factors determining their distribution within patches, deer should be equally selective in food patches of different shapes. However, when foraging collectively, fallow deer tend to form rounded groups in exposed areas, possibly in response to predation risk (Focardi & Pecchioli, 2005). We therefore hypothesized that selecting the highest quality food as a cohesive group would be easier in block rather than line patches.

Information transmission may also play a role in how the distribution of food quality within patches influences diet selectivity. We manipulated food quality within patches by applying a tannin-rich plant extract to the food pellets; tannins are plant secondary metabolites that reduce palatability and hence food preference (Bergvall, Rautio, Luotola, & Leimar, 2007; Bernays, Driver, & Bilgener, 1989). If deer make efficient use of group information, they should consume more of the high-quality (low-tannin) food when quality differs between the two sides of a patch (i.e. a single contrast at the patch centre) than when quality alternates between neighbouring feeders. This is because the group should use shared information about the distribution of food quality to move to the

better half of the patch, which is not possible when quality alternates between neighbouring feeders. On the other hand, if the information is not shared efficiently, individuals would need to rely on personal information. We would then expect better diet selection in patches of alternate food quality because more deer would be able to directly compare low- and high-tannin food in neighbouring feeders. In single-contrast patterns, the food quality contrast between neighbouring feeders only exists at the centre of the patch. We also compared food patches where low-tannin food was located centrally or peripherally in blocks to test whether a tendency by fallow deer to congregate centrally within patches was independent of food quality. If so, we predicted that diet selectivity would be weaker in patches with high-tannin than low-tannin centres. By influencing foraging efficiency, patch shape and the distribution of food quality within patches may be important constraints on the benefits of group foraging. In our study, we quantified diet selectivity at the group level under different patch treatments, allowing us to test how efficiently individuals transmitted information to group members and examine the potential for within-patch choices that are at odds with food quality.

METHODS

The Herbivore

Fallow deer are classified as generalist herbivores and intermediate opportunists, having features of both grazers and browsers depending on the season (Apollonio, Focardi, Toso, & Nacci, 1998; Hofmann, 1989). We conducted our experiment using a group of captive fallow deer kept in a 20 ha enclosure at Tovetorp zoological research station (Department of Zoology, Stockholm University) in south-central Sweden. The group consisted of 55 individuals that were mainly females with offspring and only one mature male; younger males were removed annually from the group before they reached maturity to avoid inbreeding. All the deer had ad libitum access to water, salt stone and their natural feed (grasses, pine, *Pinus sylvestris*, Norway spruce, silver birch, *Betula pendula*, aspen, *Populus tremula*, rowan, *Sorbus aucuparia*, alder, *Alnus glutinosa*, and European raspberry, *Rubus idaeus*), including during the trials, and were not fasted at any point during the study period.

Experimental Design

We simulated plant patches by creating a bioassay where both the quality and spatial arrangement of the offered food were manipulated. As artificial plants we used fodder (pellets: Avrens-pellets by Lantmännen, Sweden) made of corn, milling and sugar beet by-products, minerals, vitamins, fat and vegetable oils containing 95 g/kg of crude protein and 9.2 MJ/kg of energy. Deer select plants in a similar manner to artificial food (Bergvall et al., 2008). We manipulated food quality by adding Quebracho extract (Unitán Saica, Buenos Aires, Argentina; condensed tannin concentration 76–78%) to pellets. In general, tannins have a bitter and astringent taste and decrease palatability (Bernays et al., 1989; Robbins, Mole, Hagerman, & Hanley, 1987). The astringency correlates with the total content of tannins and can thus serve as a reliable cue to foraging animals to avoid tannin-rich plant material (Mali & Borges, 2003; Prinz & Lucas, 2000). Applying Quebracho to food pellets reduced intake by fallow deer in previous research (Bergvall et al., 2007).

The tannin was dissolved in water and sprayed on pellets, making the concentrations of condensed tannins in pellet dry mass either 0.3% or 1.5% (in the following, these percentages will be referred to as low- and high-tannin food, respectively). After spraying, pellets were allowed to dry overnight at room temperature before being used in feeding trials. The range of tannin levels used here was within the range found naturally in tree foliage used by deer. For example, levels of condensed tannins have been found to vary in oak (*Quercus*) leaves from close to 0% up to 20% of dry mass (Forkner, Marquis, & Lill, 2004; Salminen et al., 2004).

Food was offered in feeders (12-litre buckets), placed in wooden support structures to prevent the deer overturning them. We created two types of patch shape, line and block, by positioning 24 feeders in either a single row or a quadrilateral block with three rows and eight columns (Fig. 1). Based on pilot trials, the distance between feeders was selected so that it allowed three to four deer to feed from each feeder in both line and block patches. This resulted in feeders 1 m apart in lines and 2 m in blocks. Hence the line patches were 23 m long and block patches were 4 m wide and 14 m long. In each patch, 12 feeders contained low-tannin food (pellets) and 12 contained high-tannin food, with 2000 g of food in each feeder. The positions of the low- and high-tannin feeders in these set-ups were transposed between trials to prevent the deer from learning the position of the low-tannin food. We located patches in an open grass-covered area at the southern end of the enclosure.

Deer moved freely inside the enclosure and a trial was thus started when the first deer began to feed from a feeder. We allowed the herd to feed for a maximum of 20 min, at which time we disrupted the herd to end the trial and weigh the food remaining in each feeder. We selected the maximum time based on previous

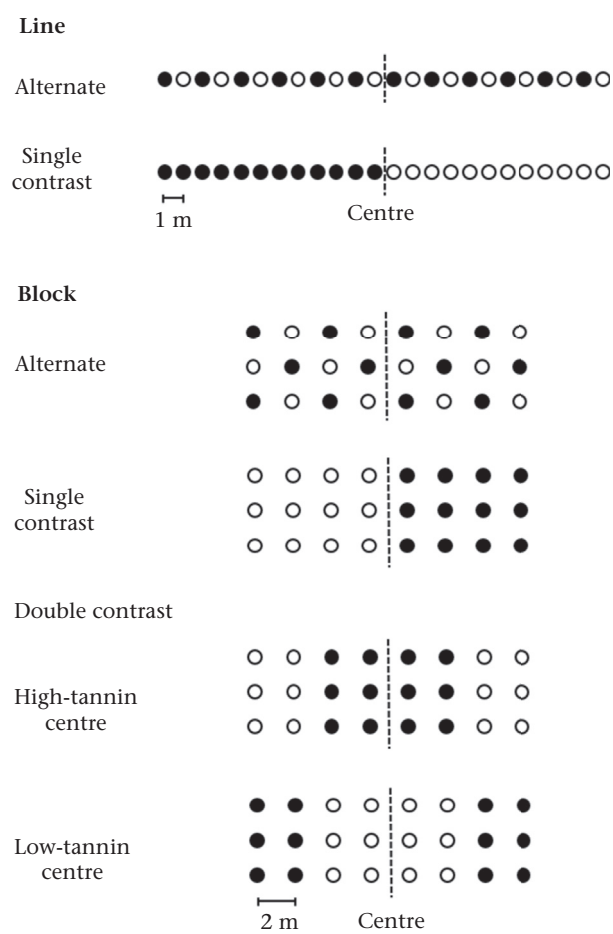

Figure 1. Schematic diagram showing the pattern of high-tannin (black) and low-tannin (white) feeders in each of the six treatments tested. The orientation of the patches was randomized.

experiments with the same herd (Rautio, Kesti, Bergvall, Tuomi, & Leimar, 2008). In a pilot study, we observed that older deer made their feeding choices quickly and sometimes returned to a patch after a period of lying down nearby; we wanted to test feeding responses to our experimental conditions and therefore limited the trial period to avoid repeat visits to already depleted patches. The length of trials ranged from 12.2 to 20.0 min (mean ± SE = 19.1 ± 0.3 min). Deer were free to enter and leave patches, and the maximum number of deer participating in each trial ranged from 26 to 33 individuals (mean ± SE = 31.9 ± 0.2).

Consumption (per feeder) was computed as the difference between the weight (g) of pellets before and after the trial. Spillage of feed was insignificant due to the large buckets and the wooden supports. We conducted trials in July and August 2003, but only during dry weather to avoid pellet weight changes from rain. We ran a maximum of two trials per day (one in the morning and the other in the afternoon). Each of the six treatments was repeated eight times in random order, resulting in a total of 48 trials and 1152 individual feeder records.

Ethical Note

We performed our study with permission from the Swedish National Board for Laboratory Animals (Dnr 14-14). Animals were free to choose whether to participate in trials and could leave the trial area at any time. Tannin levels in the pelleted food were within the range found in natural forage and we did not detect any adverse effects of its consumption.

Statistical Analyses

We expected that absolute consumption would be subject to the number of deer participating in a trial, the amount of time the herd spent in the patch and satiation, which might vary with the consumption of plants within the enclosure and whether the trial was the first or second of the day. To ensure that individual trials were comparable, we computed relative consumption by dividing the absolute consumption per feeder by the average consumption of all feeders in a trial. Hence, values below 1 indicate smaller, and values over 1 larger, consumption than on average. The possibility remained that trial times or deer numbers could affect the consumption of low- and high-tannin food differently, altering their relative consumption. To test the effects of number of deer taking part in a trial and trial time on the relative consumption of low- and high-tannin food, we fitted linear mixed-effects models including trial as a random effect. Relative consumption of low- and high-tannin food was not affected by deer number (estimate $_{\text{low-tannin}} = 0.014 \pm 0.016$, $t_{1149} = 0.83$, $P = 0.41$). The longer the trial, the more low-tannin food the deer consumed, but the increase in relative consumption per additional deer was very small (estimate $_{\text{low-tannin}} \pm \text{SE} = 0.022 \pm 0.010$, $t_{1149} = 2.13$, $P = 0.033$). Additionally, trial times did not vary significantly between treatments (Welch's $F_{5,\ 19} = 1.33$, $P = 0.29$), and thus any minor effects of trial time on relative consumption would not influence our comparisons of patch treatments.

To determine whether the relative consumption by fallow deer was affected by patch shape, spatial pattern of food quality and tannin content, we used planned contrasts in a linear mixed-effects model including trial as a random effect. We used a hierarchical, balanced contrasts matrix to test for differences in relative consumption between low- and high-tannin food, and in relative consumption of low-tannin food in line and block patches, alternate and coarser contrasts in food quality, single- and double-contrasts in blocks, and the position of high- and low-tannin food in double-contrast blocks.

We observed that deer tended to congregate around the centre of patches and were thus interested in how the distance between a given feeder and the centre of the patch affected relative consumption in the six patch treatments. First, we calculated the distance of each feeder from the centre line of each patch (Fig. 1). Feeders in alternate and single-contrast lines were located 0.5–11.5 m from the centre, while those in blocks were located 1–7 m from the centre (Fig. 1). In double-contrast blocks, feeders were also located 1–7 m from the centre but the high- and low-tannin feeders were not distributed across the full range of distances in either the high- or low-tannin centred patches, precluding direct comparison with the alternate and single-contrast blocks. We therefore constructed three separate models comparing (1) alternate and single-contrast lines, (2) alternate and single-contrast blocks and (3) double-contrast blocks with high- and low-tannin centres. For each comparison, we fitted a linear mixed-effects model testing the effects of the four combinations of quality contrast and tannin content (e.g. for lines, 'alternate: low tannin', 'alternate: high tannin', 'single contrast: low tannin', 'single contrast: high tannin'), the distance of feeders to the patch centre and the interaction of both on the relative consumption. We included trial as a random effect. The global models for all three comparisons indicated significant interactions between distance to the centre and the combined contrast by tannin factor ($\chi^2_3 > 16$, $P < 0.002$; Appendix Table A1). We thus proceeded to pairwise comparisons of slopes (relative consumption over distance to centre) for the four contrast by tannin levels in each model. *P* values were adjusted for multiple comparisons using the Tukey method.

We conducted all analyses in R statistical environment v. 3.2.4 (R Core Team, 2016). We fitted linear mixed-effects models using the 'lmer' function in the 'lme4' package (Bates, Maechler, Bolker, & Walker, 2015), run together with the 'lmtest' package to derive *P* values. To conduct pairwise comparisons of the consumption over distance slopes, we used the 'lstrends' function in the 'lsmeans' package (Lenth, 2016). The data used in these models met the assumptions of normality and homoscedasticity of residuals, as determined by visual inspection of diagnostic plots. In assessing trial times between patch treatments, the assumptions were not met and transformation was ineffective; we therefore used the Welch's heteroscedastic *F* test ('welch.test' function in 'onewaytests' package). The threshold for significance was $\alpha = 0.05$.

RESULTS

The fallow deer moved as a group and performed a similar ritual during each trial. They approached the patch in single file, with the leading individual selecting a central bowl and other individuals filling in to form a rounded group focused at the centre of the patch (see Appendix Fig. A1). During the trials, a few individuals broke away from the core group to explore more peripheral feeders. Patch departure was usually initiated by a single individual beginning to move away from the patch; individuals in close proximity then followed and the whole group eventually moved on together.

Patch Shape and Food Quality Contrasts

The mean amount of food consumed per feeder constituted less than half of that available (mean ± SE = 837 ± 19 g/feeder); of this, 71% was low-tannin food ($\text{mean}_{\text{low-tannin}} \pm \text{SE} = 1191 \pm 28$ g/feeder; $\text{mean}_{\text{high-tannin}} \pm \text{SE} = 483 \pm 17$ g/feeder). Owing to variable total consumption between trials, we analysed and describe results hereafter in terms of consumption relative to trial means. The deer consumed more low- than high-tannin food relative to trial means (Table 1, Fig. 2). They consumed more low-tannin food from feeders in block rather than line patches. In line patches, the deer

Table 1
Hierarchical planned comparisons testing effects of tannin content, patch shape and pattern of quality contrasts within patches on pellet consumption (relative to trial means) by fallow deer

Contrast			Estimate	SE	t_{1145}	P
(Intercept)			1.00	0.019		
Low tannin vs high tannin			0.429	0.019	22.27	**<0.001**
Low tannin	Line vs block		−0.060	0.019	−3.14	**0.002**
	Line	Alternate vs single contrast	0.101	0.047	2.14	**0.033**
	Block	Alternate vs single & double contrast	−0.030	0.019	−1.57	0.116
		Double contrast vs single contrast	−0.022	0.027	−0.82	0.413
		Double contrast: high-tannin centre vs low-tannin centre	−0.119	0.047	−2.53	**0.012**

Trial was included as a random effect. Significant differences are given in bold ($P < 0.05$).

consumed more low-tannin food in alternate than single contrasts in food quality. In block patches, there was no significant difference in consumption of low-tannin food between alternate and coarser (single and double) contrast patterns, nor between single and double contrasts. However, in double-contrast blocks, deer consumed less low-tannin food when the high- rather than low-tannin food was located at the centre (Table 1, Fig. 2).

Distance from the Centre of a Patch

The deer tended to consume less from feeders at greater distances from the centre in both line and block patches, with the slope of this relationship dependent on the spatial pattern of low- and high-tannin food (Table 2, Fig. 3; see also Appendix Table A1, Figs A2–A4). Consumption of low-tannin food from feeders in single-contrast lines decreased more rapidly with distance from the centre than from feeders in alternate lines, while consumption of high-tannin food decreased at a similar slower rate for both alternate and single-contrast lines (Fig. 3a). In blocks, consumption of low- and high-tannin food decreased at a similar rate with distance from the centre in both alternate and single contrasts, except for low-tannin food in the alternate pattern which decreased more rapidly than high-tannin food in both quality contrasts (Fig. 3b). In alternate blocks, deer also consumed more high-tannin food from central feeders than low-tannin food from peripheral feeders on average; this did not occur in single-contrast blocks. In double-contrast blocks, the rate of decrease in consumption with distance from the centre was greater for low-tannin food at the edges than for high-tannin food at both the edges and the centre (Fig. 3c).

DISCUSSION

By experimentally manipulating the quality and spatial arrangement of food, we demonstrated that diet selectivity in a group of social herbivores was influenced by both the spatial characteristics of the food patch and the cohesive behaviour of the animals. We reject the hypothesis that group-foraging herbivores efficiently transmit information about the distribution of food quality in a patch. Instead, we suggest that poor diet selection can result from the inability of group members to communicate efficiently about within-patch spatial variation in food quality, and from factors unrelated to food quality that affect movement decisions. These processes limit foraging efficiency and therefore constrain the benefits of foraging in a group.

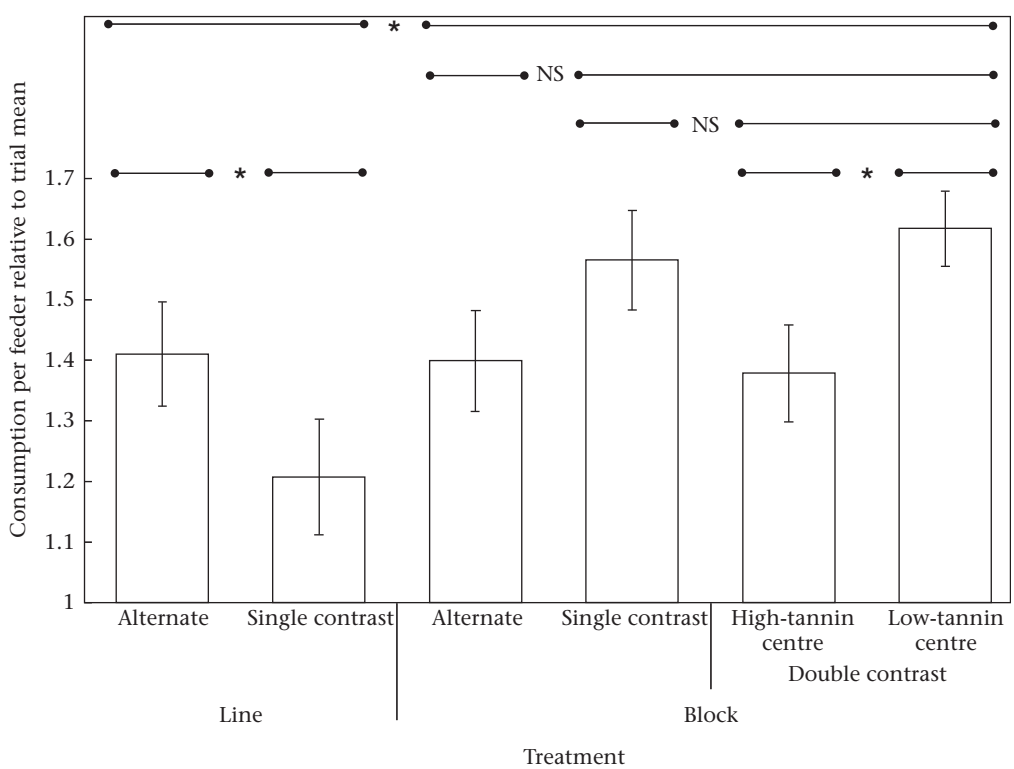

Figure 2. Amount of low-tannin pellets that fallow deer consumed per feeder relative to mean consumption for the trial (mean ± SE), by patch shape and food quality contrast within the patch. Planned comparisons are shown as black horizontal lines. *$P < 0.05$ (statistical details are in Table 1).

Table 2
The effect of distance from patch centre on the relative consumption of high- and low-tannin food by fallow deer, compared between patch treatments

Contrast	Estimate	SE	t_{384}	P
Line patches (alternate vs single contrast)				
A: high tannin vs A: low tannin	0.090	0.018	4.967	**<0.0001**
A: high tannin vs SC: high tannin	0.028	0.018	1.555	0.406
A: high tannin vs SC: low tannin	0.138	0.018	7.605	**<0.0001**
A: low tannin vs SC: low tannin	0.048	0.018	2.638	**0.043**
SC: high tannin vs A: low tannin	0.062	0.018	3.412	**0.004**
SC: high tannin vs SC: low tannin	0.110	0.018	6.050	**<0.0001**
Block patches (alternate vs single contrast)				
A: high tannin vs A: low tannin	0.120	0.034	3.559	**0.002**
A: high tannin vs SC: high tannin	0.009	0.034	0.269	0.993
A: high tannin vs SC: low tannin	0.075	0.034	2.225	0.118
A: low tannin vs SC: low tannin	−0.045	0.034	−1.334	0.542
SC: high tannin vs A: low tannin	0.111	0.034	3.291	**0.006**
SC: high tannin vs SC: low tannin	0.066	0.034	1.957	0.206
Block patches (double-contrast: high- vs low-tannin centre)				
HTC: high tannin vs HTC: low tannin	0.276	0.073	3.759	**0.001**
HTC: high tannin vs LTC: high tannin	0.059	0.073	0.800	0.854
HTC: high tannin vs LTC: low tannin	0.116	0.073	1.578	0.392
HTC: low tannin vs LTC: low tannin	−0.160	0.073	−2.181	0.130
LTC: high tannin vs HTC: low tannin	0.217	0.073	2.959	**0.017**
LTC: high tannin vs LTC: low tannin	0.057	0.073	0.778	0.865

A = alternate, SC = single contrast, HTC = high-tannin centre, LTC = low-tannin centre. Significant differences are given in bold.

Overall, fallow deer consumed more low- than high-tannin food, consistent with a significant body of previous research involving tannin consumption in this species (Alm, Birgersson, & Leimar, 2002; Bergvall, 2009; Bergvall et al., 2008) and several other mammalian herbivores (McArthur & Sanson, 1993a, 1993b; Provenza et al., 1990). The group level preference for low-tannin food confirmed our experimental premise that tannin content was negatively related to food quality as perceived by the deer. Importantly, the level of selectivity for low-tannin food differed in response to both patch shape and the spatial pattern of food quality within patches. We suggest that these differences were in part driven by social cohesion as deer tended to cluster at the centre of patches. While the same amount of high- and low-tannin food was available in all treatments, cohesive behaviour restricted exploration within patches, altering the effective availability of food qualities as a function of patch spatial characteristics. This reduced the quality of the diet consumed by the group in some patch treatments, suggesting that the perceived benefits of group cohesion outweighed those of diet selectivity. These novel findings extend our understanding of the trade-offs made by social herbivores foraging in heterogeneous resource patches.

Deer made better diet decisions in block than line patches, consuming more low-tannin food in the former than the latter. We suggest that this was a result of cohesive behaviour that reduced the accessibility of good-quality food further away from the centre of a patch. While the same amount of high- and low-tannin food was available in both block and line patches, feeders were not equally accessible given that individuals concentrated foraging efforts at the centre of a patch. In all treatments (except double-contrast blocks with high-tannin centres), consumption was greatest at the centre of the patch and decreased towards the edges. There were six feeders available within 1 m of the centre in block patches compared to only two feeders in line patches, with each feeder taking the deer further away from the centre of the patch and the group. Thus, the desire of individuals to be as close as possible to the core group at the centre of a patch probably led to poorer diet selection when food was presented in lines compared to blocks. We suggest that the deer traded off diet selectivity in favour of reduced predation risk in line patches (Beecham & Farnsworth, 1999).

In line patches, deer selected a better diet when presented with alternate than single-contrast patterns of food quality. This was in contrast to our expectation that the group would simply move to the low-tannin side of single-contrast lines, maintaining group cohesion while accessing the best food; instead, they maintained the highest consumption pressure at the centre. In contrast, sheep consume more preferred food when it is spatially aggregated rather than randomly dispersed in small clumps (Clarke, Welch, & Gordon, 1995; Dumont, Maillard, & Petit, 2000; Edwards, Newman, Parsons, & Krebs, 1994). It appears that fallow deer in our study were too inflexible to adjust the group position in the patch to coincide with the higher quality food. We suggest that the aggregation at the centre of patches may come about from the interdependent micro-movements of individuals once in a patch, as they reposition themselves relative to neighbours to try to reduce their individual predation risk while still accessing forage. This effect may have disappeared over time if the deer had repeatedly encountered singe-contrast patches because they may have learnt the spatial pattern of food within the patch, as seen by Edwards et al. (1994) in sheep.

The consumption of low-tannin food decreased more rapidly towards the edges of lines in single-contrast than alternate patterns of food quality. This may have occurred because most of the deer foraging in single-contrast lines could not easily compare high- and low-tannin food. Fallow deer are known to adjust consumption decisions, including tannin intake, based on previous experiences; Bergvall et al. (2007) found that fallow deer consumed fewer high-tannin pellets when they shifted from another bowl of low- rather than high-tannin pellets. Thus in alternate lines, experience with nearby low-tannin feeders may have reduced consumption of high-tannin food, while in single-contrast lines, only deer at the centre may have experienced the quality contrast. In the latter, the few individuals that were informed about the relative quality of the food did not transmit this information to their peers that were unable to make comparisons. Additionally, deer at the periphery of groups can feed more efficiently due to decreased conspecific interference (Focardi & Pecchioli, 2005), and this interference would probably be lowest at the outer edges of the high-tannin food.

Diet selectivity by the deer was not significantly influenced by the frequency of quality contrasts in a block patch. In blocks, there was no significant difference between alternate and single or double contrasts, nor between single and double contrasts in food quality. We suggest that circular groups could still make good choices between high- and low-tannin foods in these more spatially compact patches. However, there was an effect when the socially driven distribution of animals coincided with the spatial distribution of food quality within the patch. The deer made poorer diet selections when high- rather than low-tannin food was located in the middle of double-contrast blocks, consuming less low-tannin food in the former. Despite poorer selectivity, deer consumed more low-tannin food even when the high-tannin food was at the centre. This suggests that, in the middle of patches with high-tannin centres, there was a reduction in either intake rate, since fallow deer generally eat more slowly from less preferred foods (Bergvall & Leimar, 2005), or cohesion. Deer consumed the most food from low-tannin feeders closest to the centre (where they contrasted with the central high-tannin feeders) but consumption rapidly declined towards the edges, with low-tannin food at the edges of patches being underexploited. Thus, in double-contrast patches, where food quality differed significantly between the centre and edges, the within-patch pattern of food quality either reinforced social preferences for the centre of patches or resulted in elevated consumption of poor (high-tannin) food. Regardless of the mechanism resulting in aggregation at patch centres, our results clearly

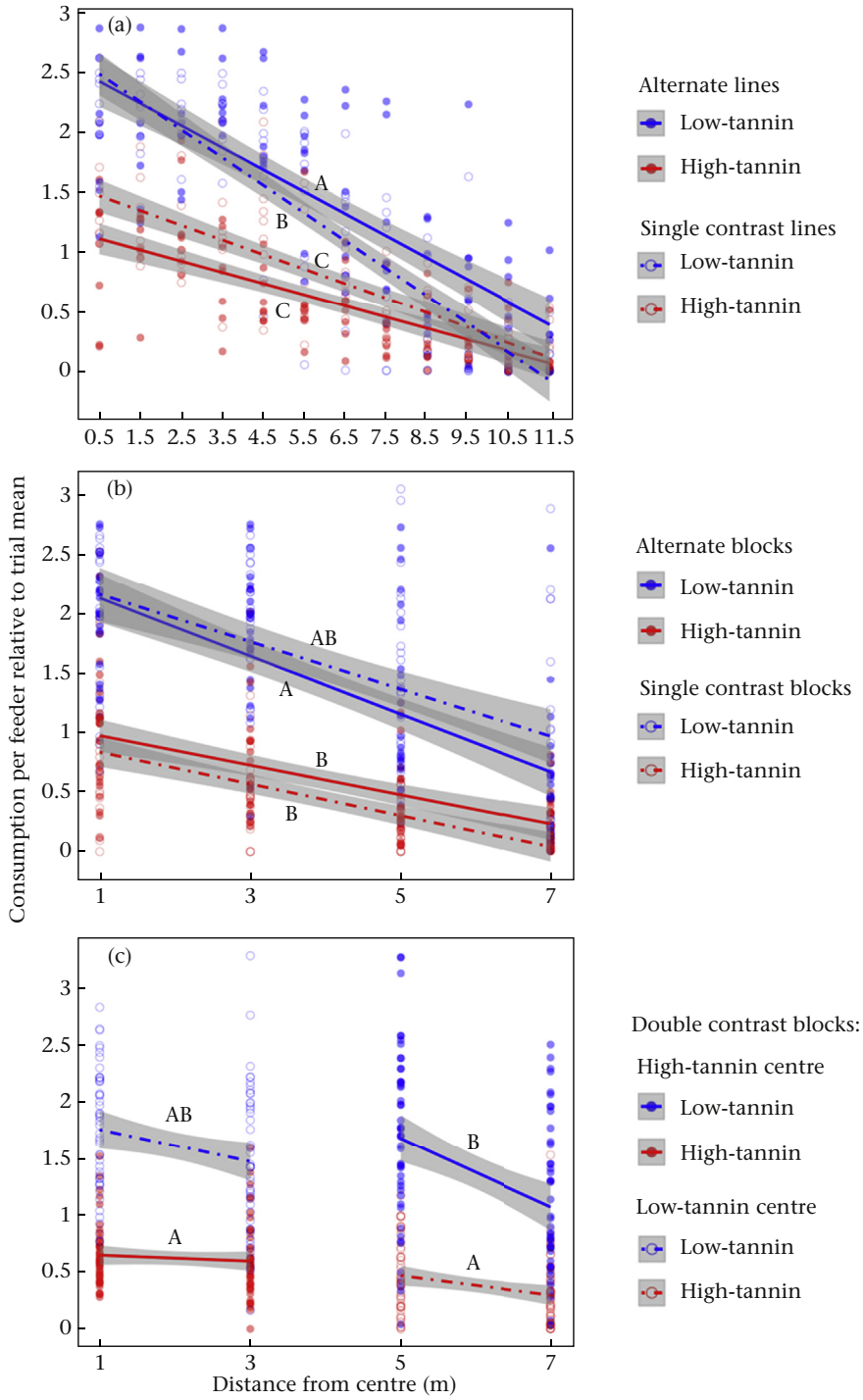

Figure 3. Linear regression of relative pellet consumption as a function of distance from the centre of the patch for each food quality contrast: (a) alternate compared to single contrasts in line patches, (b) alternate compared to single contrasts in block patches and (c) double contrasts with low- or high-tannin at the centre in block patches. Capital letters denote significant differences in the interaction (i.e. slope) between treatment and distance from the centre of the patch. Grey shading represents 95% confidence intervals.

demonstrate suboptimal diet selectivity resulting from social requirements. Like fallow deer, sheep have been shown to be reluctant to move away from the core group to reach preferred food, but the desire to stay with a group can sometimes be overridden by dietary preferences (Dumont & Boissy, 2000; Scott et al., 1995).

Suboptimal diet selection does not equate to a suboptimal foraging strategy, as this is subject to many factors other than just food quality. Indeed, selection of lower quality food is not necessarily suboptimal as part of a broader foraging strategy where a certain quantity of food must be ingested (Stutz et al., 2017). Increased intraspecific competition may manifest itself in ingestion of a lower quality but higher quantity diet to achieve nutritional requirements (Banks, Hume, & Crowe, 1999). In our study, deer had access to 24 kg of low-tannin food in each trial but never consumed it all, supporting the inference that poorer diet selection was not solely the result of competition. Ours is the first study to illustrate

that social cohesion within a patch with abundant resources can drive suboptimal diet selection by a large herbivore.

In experiments with sheep, it has been shown that witnessing the movement of a herd member, independent of rank, age and sex, is enough to initiate movement in others (Pillot et al., 2010). Therefore, reaction to movements seems to be innate and not a decision made after considering why the conspecific is moving. We suggest that foraging decisions in gregarious species that form ephemeral groups are influenced more by their desire to be close to conspecifics than by the distribution of food. The formation of rounded groups of ungulates in open habitats has been noted in both fallow deer (Focardi & Pecchioli, 2005) and forest buffalo, *Syncerus caffer nanus* (Melletti, Delgado, Penteriani, Mirabile, & Boitani, 2010). This may be a result of individuals seeking protection from predators at the centre of the group (Hamilton, 1971), at the cost of foraging efficiency (Beecham & Farnsworth, 1999). Although synchronization involves costs, benefits from aggregation such as protection from predation might reimburse such costs. Free-ranging herbivores experience greater hunger and exposure to predators than our captive herd of fallow deer; we therefore expect that diet selectivity would be sacrificed to an even greater extent in the wild to both maximize quantitative ingestion and reduce predation risk. This begs the question: where is the tipping point in food quality contrasts that would lead herbivores to prioritize diet selection over social cohesion?

Here, we tested patch characteristics affecting diet selection at the group level. However, diet selectivity could vary significantly between individual fallow deer foraging in a group. In contrast to foragers that regularly change positions within the group (Krause, 1993), fallow deer exhibit fairly static positions relative to other herd members (Focardi & Pecchioli, 2005). This means that individual deer that tend to take central positions in the herd could face different dietary choices to peripheral individuals, depending on the distribution of food quality within the patch. The consequences of within-patch patterns in food quality could thus differ substantially between individuals, with the potential to ultimately affect fitness (Hamilton, 1971; Krause, 1994). Experiments that can assess intake by individuals in a group could be used to clarify whether there are differences in costs and benefits based on position within the group.

Shared information about the location and quality of food can improve foraging efficiency for animals foraging in groups rather than alone. However, our study illustrates two mechanisms that limit diet selectivity in groups: (1) restricted transmission of information between group members and (2) movement decisions based on factors external to food acquisition. Fallow deer selected better diets when more individuals directly experienced the quality contrast, suggesting that even when foraging in a group, herbivores may rely heavily on personal information about food quality. Group foragers may prioritize other critical needs such as the management of predation risk and these could be key to driving collective movement decisions. Quantifying the influence of competing needs on movement decisions is fundamental to understanding collective behaviour.

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DATA ACCESSIBILITY

All data are available through Mendeley Data (https://doi.org/10.17632/7hhd22c8yx.1).

Supplementary material

Supplementary material associated with this article can be found at https://doi.org/10.1016/j.anbehav.2017.11.004.

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Appendix

Table A1
Global tests of effects on relative consumption by fallow deer: quality contrasts in low- and high-tannin food and the distance from patch centre

Patch shape	Fixed effect	X^2	*df*	*P*
Line (alternate, single contrast)	Quality contrast by tannin content	214.697	3	**<0.0001**
	Distance from centre	611.21	1	**<0.0001**
	Interaction	70.057	3	**<0.0001**
Block (alternate, single contrast)	Quality contrast by tannin content	336.405	3	**<0.0001**
	Distance from centre	216.408	1	**<0.0001**
	Interaction	17.065	3	**<0.001**
Block (high-, low-tannin centre)	Quality contrast by tannin content	375.117	3	**<0.0001**
	Distance from centre	28.683	1	**<0.0001**
	Interaction	15.687	3	**0.001**

Alternate and single contrasts in food quality are compared within line and block patches, and high- and low-tannin centres are compared within block patches only (three separate models). Significant differences are given in bold ($P < 0.05$).

Figure A1. Typical formation of fallow deer feeding from (a) line and (b) block patches during trials.

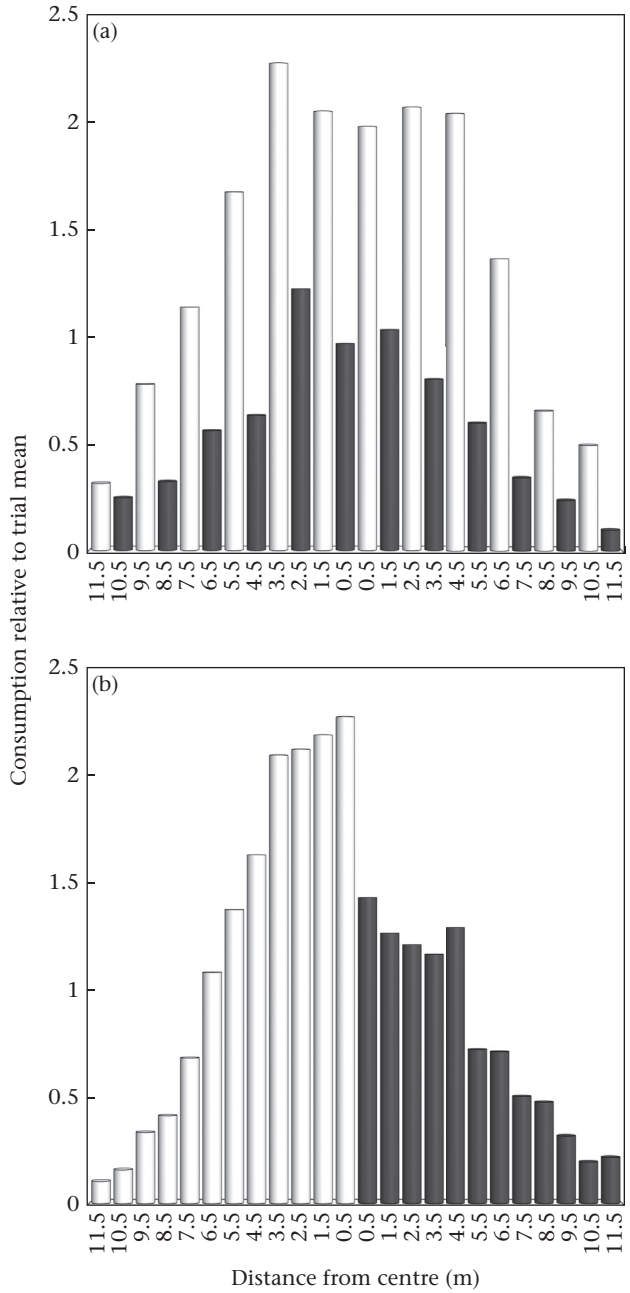

Figure A2. Mean pellet consumption (relative to trial means) by fallow deer at each feeder position in line patches with (a) alternate and (b) single contrasts in food quality. White bars represent feeders with low-tannin (high-quality) pellets and black bars represent feeders with high-tannin (low-quality) pellets.

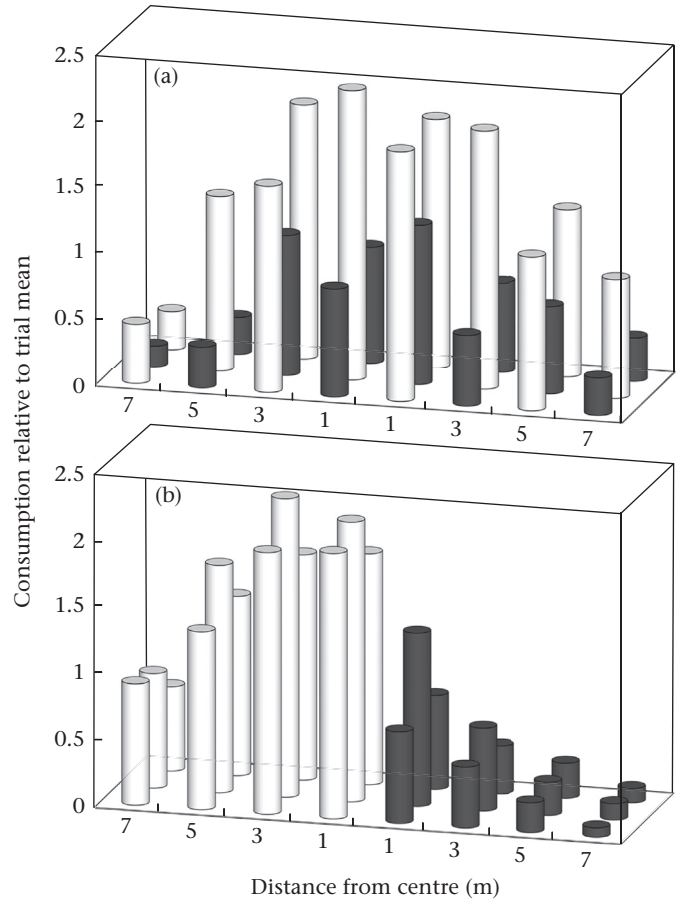

Figure A3. Mean consumption (relative to trial means) by fallow deer at each feeder position in block patches with (a) alternate and (b) single contrasts in food quality. White bars represent feeders with low-tannin (high-quality) pellets and black bars represent feeders with high-tannin (low-quality) pellets.

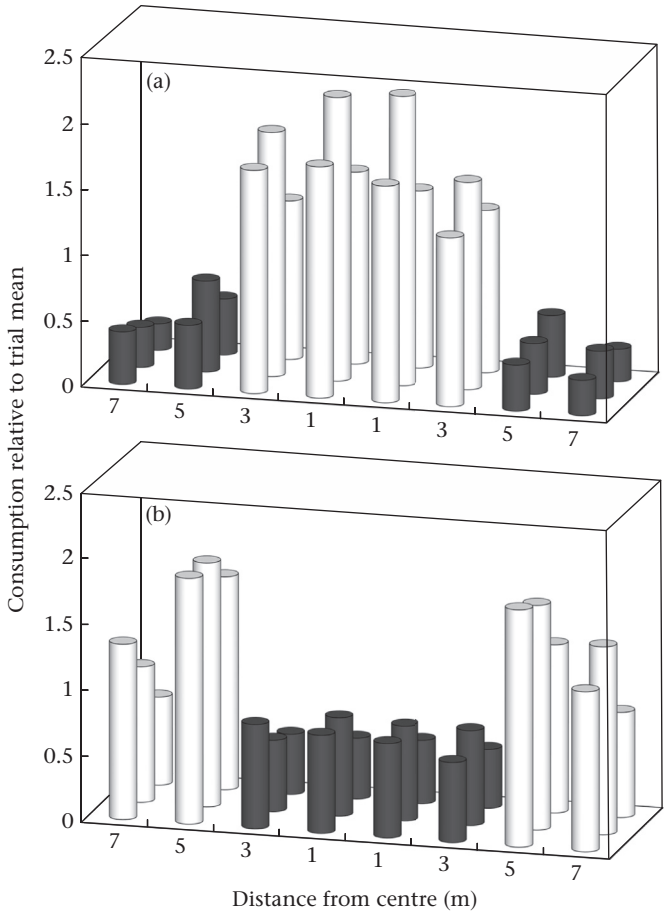

Figure A4. Mean consumption (relative to trial means) by fallow deer at each feeder position in double-contrast block patches with (a) low-tannin centres and (b) high-tannin centres. White bars represent feeders with low-tannin (high-quality) pellets and black bars represent feeders with high-tannin (low-quality) pellets.